End-to-End Geometric Canonicalization for 2D Chest X-Ray Preprocessing and Downstream Robustness

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***Abstract*—Inter-patient anatomical variation, posture shifts, and framing differences in 2D chest radiographs add input noise that degrades the performance and generalization of downstream computer-aided diagnosis (CAD) models. Common preprocessing steps such as heuristic resizing or rigid cropping do not correct rotational, scaling, or structural misalignment. We propose a modular geometric canonicalization pipeline that maps raw radiographs into a fixed-resolution canonical frame. The pipeline performs horizontal alignment by symmetry minimization, locates the lungs with a pre-trained U-Net segmentation model, and applies an aspect-preserving closed-form affine transformation so that the lungs occupy a fixed fraction of the frame height. A custom intensity transform is used to strengthen anatomical edges before localization. We evaluate the pipeline with a ResNet-18 classifier trained on 1,321 RSNA chest radiographs with and without preprocessing. Preprocessing raised validation sensitivity in early-to-mid training (about 0.75 compared with 0.55 to 0.62 without it), but reduced and destabilized specificity, and the validation AUC stayed broadly similar across both settings. We also analyze failure cases caused by rotation and low contrast, which point to the additional preprocessing layers needed for robust deployment.**

***Index Terms*—chest radiography, geometric canonicalization, lung segmentation, U-Net, image registration, computer-aided diagnosis, preprocessing**

I. Introduction

Disease and anomaly detection on chest radiographs with deep learning is a key application of AI in medicine. Model training benefits from a well-structured dataset, and the quality of that dataset strongly affects how quickly and how accurately a model can be trained. Radiographs are acquired from a wide range of patients, including infants and adults over 60, whose body shapes deviate from an idealized thoracic outline. The lungs of an infant occupy a much smaller fraction of the frame than those of an adult, and the subject may be slightly off-center or rotated during acquisition. Each of these small geometric errors compounds during training.

Conventional preprocessing for this problem relies on heuristic resizing or rigid cropping. Such operations do not correct rotation, scale, or structural misalignment, so the network must learn to tolerate these variations from the data itself. This paper proposes a preprocessing module that aligns 2D radiographs to a standard geometric frame. The goal is to reduce input variability and improve training for downstream models. The contributions are as follows:

- A multi-stage canonicalization pipeline that decouples anatomical localization, aspect-preserving geometric registration, and quality verification.
- A symmetry-based horizontal alignment procedure that does not require binarization.
- A closed-form affine normalization that places the lungs at a fixed proportion of the frame height and records the transformation parameters for traceability.
- An empirical comparison of a ResNet-18 classifier trained with and without the pipeline, together with an analysis of its failure modes.

II. Background and Inspiration

The design takes its inspiration from MediaPipe Face Mesh, a machine learning system developed by Google for real-time facial landmark detection. It estimates 468 three-dimensional landmarks from a single camera without a depth sensor, first detecting the face and then estimating the position of each landmark to form a dense mesh [7] (Fig. 1). We adopt the same two-stage logic: a coarse detector localizes the anatomy of interest, and a finer geometric model then normalizes its pose and scale.

Lung segmentation in chest radiographs has been studied with anatomical atlases and nonrigid registration [1], and public radiograph databases with and without lung nodules are widely used for evaluation [2]. For this work, rather than training a new segmentation network, we reuse a U-Net lung segmentation model trained on the Montgomery and Shenzhen frontal chest radiograph datasets [3]–[5]. These datasets were assembled for tuberculosis screening. A second public Kaggle implementation of the same model [6] was also consulted.

III. Proposed Method

The pipeline has four modules: image alignment, lung segmentation, region-of-interest extraction, and geometric normalization. Rotation correction is planned as a future step; the current implementation focuses on horizontal alignment and scaling.

A. Lung Localization Without Binarization

A thresholding-based localization was tried first. It proved unreliable for three reasons. First, radiographs differ widely in brightness, so no single threshold separates the chest from the background. Second, the cardiac silhouette skews the left lung

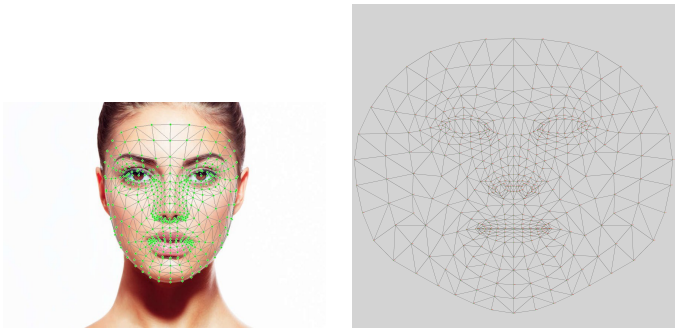
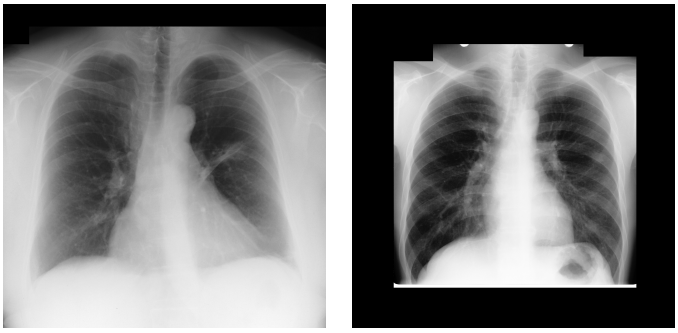


Fig. 1: Face landmark meshes from MediaPipe Face Mesh, which motivated the landmark-based design of the pipeline.



(a) Enlarged; outer edges cropped (b) Shrunk; zero padding added

Fig. 2: Scaling to a fixed lung-height ratio. (a) Lungs smaller than the target are enlarged. (b) Lungs larger than the target are reduced and zero-padded.

boundary and makes the lung–heart junction hard to delineate. Third, the lung boundaries against the lateral chest wall are not captured well after binarization. For these reasons, binarization and similar hard segmentations were removed from the pipeline before bounding-box estimation.

Edge detectors were considered next. Canny and Sobel operators assume a binary input, and radiographs are continuous grayscale images with low gradients between dark and light regions. To make the gradients more pronounced, the raw image is passed through the transform

$$I'(x) = 255 - |255 - 2I(x)|\,, \tag{1}$$

where $I(x)$ is the original grayscale intensity and $I'(x)$ is the transformed intensity. The transform was chosen empirically because it amplifies intensity transitions, which we hypothesized would make anatomical boundaries easier to detect. It was evaluated specifically for the edge-detection task on the X-ray images used in this study.

B. Horizontal Alignment

Horizontal alignment centers the subject by exploiting the approximate left–right symmetry of the thorax. The procedure is iterative. At each step, two columns of pixels are removed from the right side of the image, the image is flipped left to right, and the deviation between the image before and after flipping is measured. A single black column is then added to each side, and the search continues until the error exceeds τ times the minimum error found so far. The same search is applied to the left side.

The factor τ trades accuracy for compute time. A large value (for example 3 or 4) forces the search to examine nearly every pixel column, while a value close to 1 (for example 1.01) can stop too early and assume zero displacement. Horizontal alignment matters for two reasons. It removes dark, unwanted regions near the armpits that can confuse the U-Net and cause it to segment structures other than the lungs. It also reduces the work required by the scaling stage.

C. Scaling and Bounding Box

Radiographs may contain small subjects whose lungs sit in the upper half of the frame. To standardize the field of view, we locate the lungs with the pre-trained U-Net and build a bounding box around the segmented mask. The vertical position and scale are then normalized so that the lungs occupy a fixed fraction of the frame height. We set the target height ratio to $r = 0.75$.

Let c_x and c_y be the horizontal and vertical centers of the bounding box, and let h be its height. The scale factor is

$$s = \frac{r\,H_{\text{out}}}{h}, \tag{2}$$

where H_{out} is the output frame height. The transformation is an affine matrix that scales the image and translates it so that the lung center lands at the frame center:

$$A = \begin{bmatrix} s & 0 & \frac{W_{\text{out}}}{2} - s\,c_x \\ 0 & s & \frac{H_{\text{out}}}{2} - s\,c_y \end{bmatrix}. \tag{3}$$

Two cases arise. If the lungs are smaller than the target ratio ($s > 1$), the image is enlarged and its outer edges are cropped to fit the 2048×2048 frame. If the lungs are larger than the target ratio ($s < 1$), the image is shrunk and the remaining area is padded with zero-valued pixels to reach the standard frame size. Figure 2 shows examples of both cases.

Two practical issues arose. The first is resolution: zooming into a small region of a low-resolution image produces pixelation. Since the inputs are 2048×2048, the zoom is applied to the full-resolution image. The segmentation model was trained at 512×512, so a crop covering roughly one-sixteenth of the image area still yields reliable lung masks against the background. The second is asymmetry. The heart displaces the left lung in the frame, so the lungs are not mirror-symmetric, and the lung on the patient's left may be elongated or less regular in shape. The bounding box is therefore computed from the full segmented mask rather than from a symmetric template.

D. Pipeline Output and Traceability

The normalized image is saved for downstream tasks such as feature extraction and disease classification. The pipeline also records the alignment offsets and the affine parameters for

TABLE I: Per-image preprocessing time (seconds).

Statistic	Value
Number of rows	1,327
Mean	10.1259
Median	8.0047
Standard deviation	6.1727
Minimum	1.7087
25th percentile	5.3000
75th percentile	15.1885
90th percentile	18.6550
95th percentile	23.6082
Maximum	32.7503

TABLE II: Per-image scaling and shifting.

Statistic	Scaling	Shifting
Number of rows	1,327	1327
Mean	1.0358	-0.8459
Median	1.0157	-5
Standard deviation	0.3204	34.4086
Minimum	0.5809	-100
25th percentile	0.9225	-21
75th percentile	1.1195	19
90th percentile	1.2346	41
95th percentile	1.3256	62.3
Maximum	1.9112	100

each image, so that any output can be traced back to its source geometry. If segmentation fails to find a valid lung region, the pipeline reports the absence of a lung region instead of passing a corrupted input downstream.

IV. Experimental Setup

A. Preprocessing Cost

We measured the per-image processing time of the pipeline. Table I summarizes the distribution of these times in seconds.

The median time is about 8 seconds per image, but the long tail (95th percentile near 24 seconds) reflects the iterative horizontal alignment search, which is sensitive to image content. Reducing the tolerance τ or the search step is the main lever for trimming this cost.

B. Downstream Training

We trained a ResNet-18 classifier [8] on 1,321 chest radiographs from the RSNA dataset under two conditions. In the first, the model was trained on the raw images. In the second, every image was passed through the pipeline, and the model was trained on the 1,321 canonicalized outputs. Both conditions used the RSNA dataset [9]. Validation sensitivity, specificity, and AUC were recorded for each epoch.

V. Results and Discussion

Figures 3, 4, and 5 compare the two conditions over epochs. The main observations are as follows.

- **Sensitivity.** Preprocessing markedly improved sensitivity during early-to-mid training, reaching about 0.75 compared with 0.55 to 0.62 without preprocessing (Fig. 4).
- **Specificity.** The preprocessed model's specificity dropped sharply in several epochs before recovering. The model trained without preprocessing kept specificity consistently high, between 0.81 and 0.91 (Fig. 5).
- **AUC.** Overall validation AUC was broadly similar in both pipelines, between about 0.75 and 0.82. Preprocessing gave a slight advantage around epoch 4, then declined sharply by epoch 7 (Fig. 3).

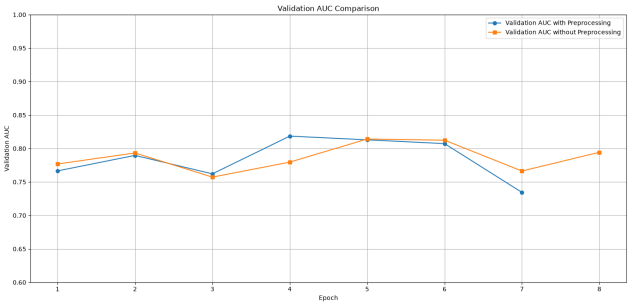


Fig. 3: Validation AUC per epoch with and without preprocessing.

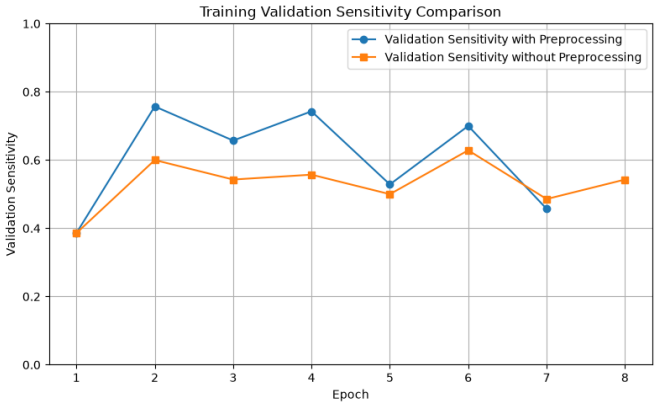


Fig. 4: Validation sensitivity per epoch with and without preprocessing.

Taken together, these results show a trade-off. Preprocessing raised sensitivity but lowered and destabilized specificity, and AUC was similar, so the benefit appears to depend on the decision threshold rather than reflecting a clear gain in overall discrimination. One possible explanation is that the pipeline was designed around the JSRT dataset, which is much cleaner than the RSNA data used for training and evaluation. Standardizing noisy RSNA images with a pipeline tuned for cleaner inputs may not transfer uniformly.

VI. Failure Cases and Limitations

Two representative failures reveal the limits of the current pipeline (Fig. 6).

In the first case, the subject was not aligned with the X-ray machine during acquisition, and the radiograph is rotated by roughly 2° clockwise. The current pipeline corrects horizontal offset and scale but does not correct rotation, which is the first of the three planned stages.

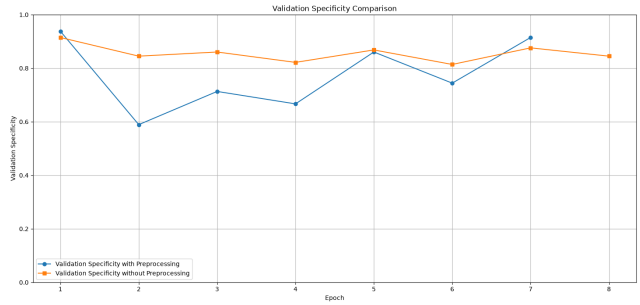
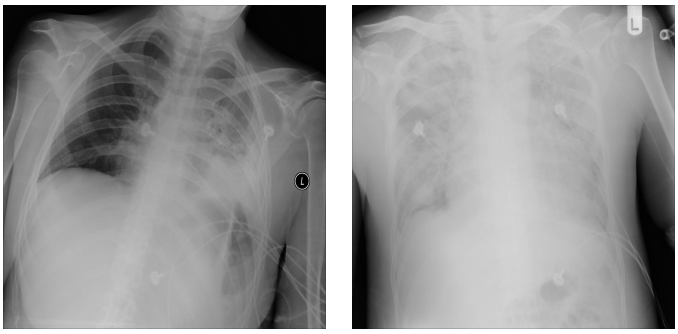


Fig. 5: Validation specificity per epoch with and without preprocessing.



(a) Rotated by roughly 20°–30°

(b) Low contrast; no lung found

Fig. 6: Failure cases of the preprocessing pipeline.

In the second case, the radiograph has too little contrast for the segmentation model to detect any lungs. The pipeline returns a "no lung found" error, which is the intended behavior for an unusable input, but it also means that such images are excluded from training. This motivates additional contrast-normalization layers before segmentation.

The pipeline is also limited by the segmentation model's training distribution, which is small and tuberculosis-focused. Its performance on pediatric radiographs, which motivated the work, has not yet been evaluated. The evaluation is also restricted to a single classifier architecture and a single dataset.

VII. Conclusion and Future Work

We presented a modular pipeline that canonicalizes chest radiographs through symmetry-based horizontal alignment, pre-trained U-Net lung localization, and an aspect-preserving affine normalization. On a ResNet-18 classifier trained on RSNA data, preprocessing improved early-to-mid training sensitivity but reduced specificity stability, and AUC was broadly unchanged. Future work will add rotation correction, contrast normalization for low-quality inputs, evaluation on pediatric and geriatric cohorts, and a systematic study of how the target lung-height ratio r and the search tolerance τ affect downstream performance.

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